

Topics:

- Vector Equations of Plane in Space,
- Distance from a Point to a Plane in Space

Equation of a Plane in Space:

A **plane in space** is determined by knowing a **point on the plane** and its “**tilt**” or **orientation**. This “**tilt**” is defined by specifying a **vector** that is **perpendicular or normal to the plane**.

Suppose that a plane M passes through a point $P_0(x_0, y_0, z_0)$ and is **normal** to the non-zero vector $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$, i.e., $\vec{n}$ is a non-zero **normal orthogonal vector**. Then M is the set of all points $P(x, y, z)$ for which $\vec{u} = \overrightarrow{P_0P}$ is **orthogonal** to $\vec{n}$. i.e., $P_0P \perp \vec{n}$. Then

$$\vec{u} = \langle x - x_0, y - y_0, z - z_0 \rangle$$

Since $\vec{n}$ is **orthogonal to the plane** thus it is **orthogonal to every vector on the plane**. The vector $\vec{u}$ lies on the plane.

Thus, the dot product

$$\vec{n} \cdot \vec{u} = 0$$

$$(a\hat{i} + b\hat{j} + c\hat{k}) \cdot [(x - x_0)\hat{i} + (y - y_0)\hat{j} + (z - z_0)\hat{k}] = 0$$

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0 \quad \text{----- (1)}$$

This equation (1) is known as the equation of plane in space.

Remark:

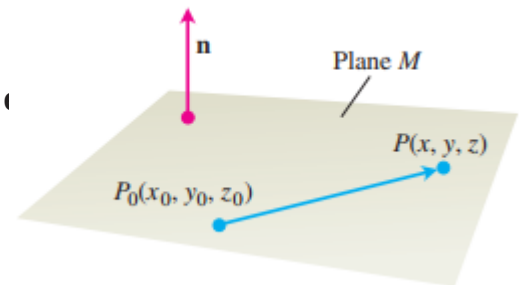
Another form of the equation of plane.

$$Ax - Ax_0 + By - By_0 + Cz - Cz_0 = 0$$

$$Ax + By + Cz - (Ax_0 + By_0 + Cz_0) = 0$$

$$Ax + By + Cz = Ax_0 + By_0 + Cz_0$$

$$Ax + By + Cz = D \text{ where } D = Ax_0 + By_0 + Cz_0$$



Example 1:

Find an equation for the plane passing through the point $P_0(-3, 0, 7)$ and perpendicular to the unit normal vector $\vec{n} = 5\hat{i} + 2\hat{j} - \hat{k}$.

Solution: Since the given point is $P_0(x_0, y_0, z_0) = P_0(-3, 0, 7)$

Tha is, $x_0 = -3, \quad y_0 = 0, \quad z_0 = 7$

And the unit normal vector is $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k} = 5\hat{i} + 2\hat{j} - \hat{k}$

Tha is, $a = 5, \quad b = 2, \quad c = -1$

Now, the equation of the plane in space is

$$a(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Putting values in the above equation, we get

$$5(x - (-3)) + 2(y - 0) + (-1)(z - 7) = 0$$

$$5(x + 3) + 2y - z + 7 = 0$$

$$5x + 15 + 2y - z + 7 = 0$$

$$5x + 2y - z + 22 = 0$$

$5x + 2y - z = -22$

Example 2: Find an equation for the plane passing through three points $A(0, 0, 1), B(2, 0, 0)$ and $C(0, 3, 0)$.

Solution: Given points are $A(0, 0, 1), B(2, 0, 0)$ and $C(0, 3, 0)$. Consider we fix a point A , then

$$\vec{AB} = \langle 2 - 0, 0 - 0, 0 - 1 \rangle = \langle 2, 0, -1 \rangle = 2\hat{i} + 0\hat{j} - \hat{k}$$

$$\vec{AC} = \langle 0 - 0, 3 - 0, 0 - 1 \rangle = \langle 0, 3, -1 \rangle = 0\hat{i} + 3\hat{j} - \hat{k}$$

Now, a vector normal to the plane is:

$$\vec{n} = \vec{AB} \times \vec{AC}$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix}$$

$$\vec{n} = \hat{i} \begin{vmatrix} 0 & -1 \\ 3 & -1 \end{vmatrix} - \hat{j} \begin{vmatrix} 2 & -1 \\ 0 & -1 \end{vmatrix} + \hat{k} \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix}$$

$$\vec{n} = \hat{i}[(0 - (-3))] - \hat{j}[-2 - 0] + \hat{k}[6 - 0]$$

$$\vec{n} = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

Now, the equation of the plane in space is

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

$$3(x - 0) + 2(y - 0) + 6(z - 1) = 0$$

$3x + 2y + 6z = 6$

Practice questions:

Thomas Calculus Ex. 12.5: 21-26

Find equations for the planes in Exercises 21–26.

21. The plane through $P_0(0, 2, -1)$ normal to $\mathbf{n} = 3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$

22. The plane through $(1, -1, 3)$ parallel to the plane

$$3x + y + z = 7$$

23. The plane through $(1, 1, -1)$, $(2, 0, 2)$, and $(0, -2, 1)$

24. The plane through $(2, 4, 5)$, $(1, 5, 7)$, and $(-1, 6, 8)$

25. The plane through $P_0(2, 4, 5)$ perpendicular to the line

$$x = 5 + t, \quad y = 1 + 3t, \quad z = 4t$$

26. The plane through $A(1, -2, 1)$ perpendicular to the vector from the origin to A

Distance from a Point to a Plane in Space

Let P be a point on the plane, and S be any point in the space, then $\vec{n}$ is the unit normal vector to the plane. Let the vector $\overrightarrow{PS}$ be denoted by the vector $\vec{u}$, i.e.

$$\overrightarrow{PS} = \vec{u}$$

Now considering the shortest distance from the point S to the plane is perpendicular to the plane, then

$$d = \frac{|\vec{u} \cdot \vec{n}|}{|\vec{n}|}$$

Proof:

From the figure, we can write

$$\cos(\theta) = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{d}{|\vec{u}|}$$

$$\cos(\theta) = \frac{d}{|\vec{u}|}$$

$$|\vec{u}| \cos(\theta) = d$$

$$d = |\vec{u}| \cos(\theta)$$

$$d = \frac{|\vec{u}| |\vec{n}| \cos(\theta)}{|\vec{n}|}$$

$$d = \frac{|\vec{u} \cdot \vec{n}|}{|\vec{n}|}$$

Hence

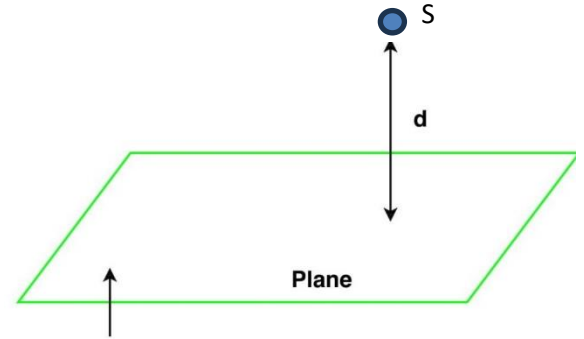


Figure 1

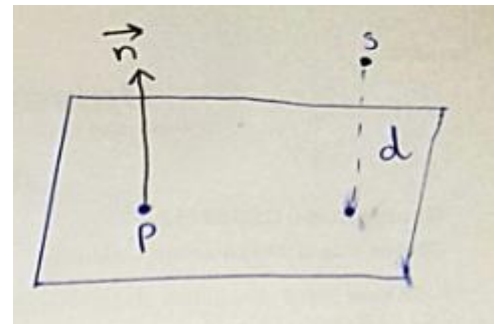


Figure 2

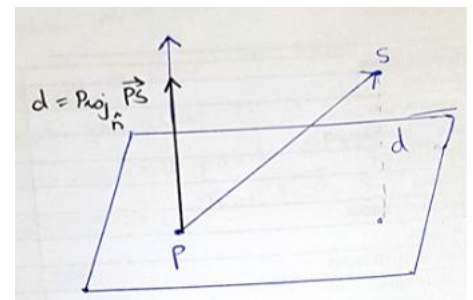


Figure 3

$$d = \frac{|\vec{u} \cdot \vec{n}|}{|\vec{n}|}$$

Example:

Find the distance from the point $S(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Solution:

The corresponding unit normal vector $\vec{n}$ to the plane is

$$\vec{n} = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

And the given point is $S = (1, 1, 3)$. We have to find the point P on the plane. i.e.,

$$P = ?$$

We find a point P in the plane and calculate the

length of the vector projection of $\vec{PS}$ onto a vector

$\vec{n}$ normal to the plane. The point on plane easiest

to find from the plane's equation are the intercepts.

To find the point P , we consider y -intercept.

That is, $(x, y, z) = (0, 3, 0)$ then

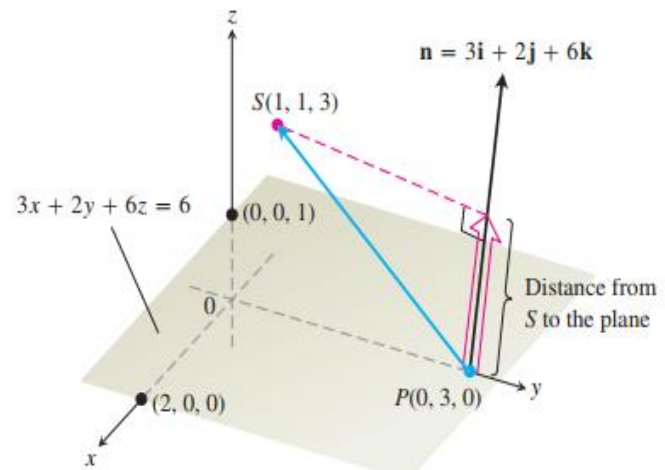
$$\begin{aligned}\vec{PS} = \vec{u} &= (1 - 0)\hat{i} + (1 - 3)\hat{j} + (3 - 0)\hat{k} \\ &= \hat{i} - 2\hat{j} + 3\hat{k}\end{aligned}$$

$$|\vec{n}| = \sqrt{(3)^2 + (2)^2 + (6)^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7$$

The distance from S to the plane is

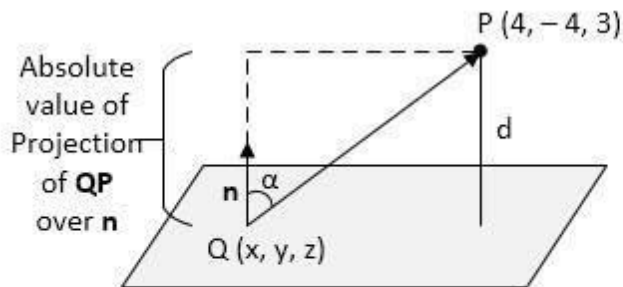
$$d = \left(\frac{|\vec{u} \cdot \vec{n}|}{|\vec{n}|} \right) = \left| (\hat{i} - 2\hat{j} + 3\hat{k}) \cdot \left(\frac{3}{7}\hat{i} + \frac{2}{7}\hat{j} + \frac{6}{7}\hat{k} \right) \right|$$

$$d = \left| \frac{3}{7} - \frac{4}{7} + \frac{18}{7} \right| = \left| \frac{17}{7} \right| = \frac{17}{7}$$



Note: We can consider x -intercept or y -intercept or z -intercept to find the point P .

Example:



Suppose the equation of the Plane is $2x - 2y + 5z + 8 = 0$ and point $P = (4, -4, 3)$. We want to find the distance (shortest) from P to the plane.

SOLUTION: From the equation of plane we know the Normal vector $\mathbf{n} = \langle 2, -2, 5 \rangle$.

Now, $\mathbf{QP} = \langle 4 - x, -4 - y, 3 - z \rangle$

Absolute value of Projection of $\mathbf{QP}$ on $\mathbf{n} = \frac{\mathbf{QP} \cdot \mathbf{n}}{\|\mathbf{n}\|}$ This is d .

$$\text{So, Distance } d = \frac{\mathbf{QP} \cdot \mathbf{n}}{\|\mathbf{n}\|} = \frac{\langle 4 - x, -4 - y, 3 - z \rangle \cdot \langle 2, -2, 5 \rangle}{\sqrt{2^2 + (-2)^2 + 5^2}}$$

We still have three unknowns. How was the distance found in the example that is demonstrated in the link that I provided?

Practice Questions:

Thomas Calculus 12 Edition: Ex. 12.5: 39-44, 45, 46

In Exercises 39–44, find the distance from the point to the plane.



39. $(2, -3, 4), \quad x + 2y + 2z = 13$

40. $(0, 0, 0), \quad 3x + 2y + 6z = 6$

41. $(0, 1, 1), \quad 4y + 3z = -12$

42. $(2, 2, 3), \quad 2x + y + 2z = 4$

43. $(0, -1, 0), \quad 2x + y + 2z = 4$

44. $(1, 0, -1), \quad -4x + y + z = 4$

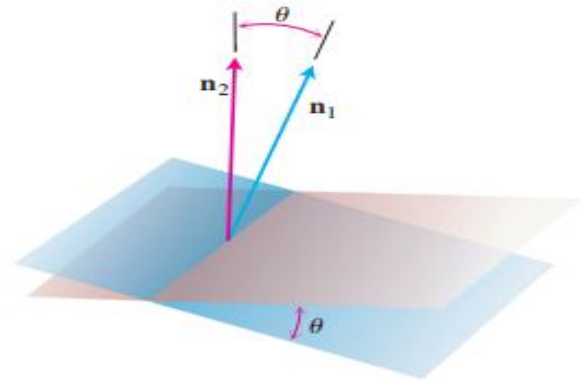
45. Find the distance from the plane $x + 2y + 6z = 1$ to the plane $x + 2y + 6z = 10$.

46. Find the distance from the line $x = 2 + t, y = 1 + t, z = -(1/2) - (1/2)t$ to the plane $x + 2y + 6z = 10$.

Angle between Two Planes

The angle between two intersecting planes is defined to be the angle between their normal vectors.

$$\theta = \cos^{-1} \left(\frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right)$$



Example:

Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Solution:

$$\vec{n}_1 = 3\hat{i} - 6\hat{j} - 2\hat{k}$$

$$\vec{n}_2 = 2\hat{i} + \hat{j} - 2\hat{k}$$

$$\vec{n}_1 \cdot \vec{n}_2 = 6 - 6 + 4$$

$$\vec{n}_1 \cdot \vec{n}_2 = 4$$

$$|\vec{n}_1| = \sqrt{(3)^2 + (-6)^2 + (-2)^2} = \sqrt{9 + 36 + 4} = \sqrt{49} = 7$$

$$|\vec{n}_2| = \sqrt{(2)^2 + (1)^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

Now

$$\theta = \cos^{-1} \left(\frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right)$$

Putting values, we get

$$\theta = \cos^{-1} \left(\frac{4}{7 \times 3} \right)$$

$$\theta = \cos^{-1} \left(\frac{4}{21} \right)$$

$$\theta = 79^\circ$$

Practice Questions:

Thomas Calculus 12 Edition Ex. 12.5: Q # 47-52

Angles

Find the angles between the planes in Exercises 47 and 48.

47. $x + y = 1, \quad 2x + y - 2z = 2$

48. $5x + y - z = 10, \quad x - 2y + 3z = -1$

Use a calculator to find the acute angles between the planes in Exercises 49–52 to the nearest hundredth of a radian.

49. $2x + 2y + 2z = 3, \quad 2x - 2y - z = 5$

50. $x + y + z = 1, \quad z = 0$ (the xy -plane)

51. $2x + 2y - z = 3, \quad x + 2y + z = 2$

52. $4y + 3z = -12, \quad 3x + 2y + 6z = 6$